

NETWORKED CONTROL

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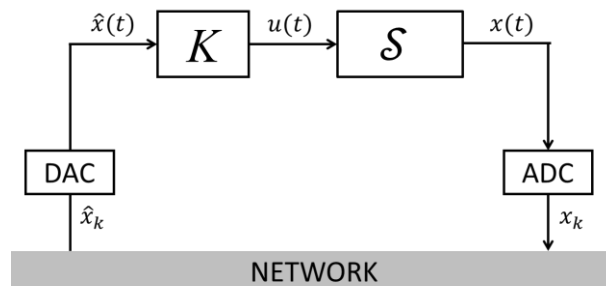
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EXERCIZES

1) Consider the continuous-time system

$$\dot{x} = 2x + 4u \quad (i)$$

- A) Answer the following questions concerning the continuous-time system (i).
- Compute the eigenvalue and the mode of the system. Is the system open-loop asymptotically stable?
 - Assume that the system is controlled using a continuous-time control law of the type $u(t) = Kx(t)$. Define the range of values of K that make the closed-loop system asymptotically stable.
- B) Compute the parameters F and G of the corresponding discrete-time dynamical equation
- $$x_{k+1} = Fx_k + Gu_k \quad (ii)$$
- with sampling time $h = \log_e 2$ s and answer the following questions concerning the discrete-time system (ii).
- Compute the eigenvalue and the mode of the system. Is the system open-loop asymptotically stable?
 - Assume that the system is controlled using a digital control law of the type $u_k = Kx_k$. Define the range of values of K that make the closed-loop system asymptotically stable.
- C) Assume that the system is controlled through the networked control system (NCS) sketched in the following figure.



We set $h = \log_e 2$ s. Answer the following questions.

- Define the range of values of K that make the closed-loop system asymptotically stable in case the network is affected by a constant delay $\tau = \log_e 1.1$ s in absence of packet dropouts?
- In case $K = -2/3$, what is the range of values of the constant delay τ such that the closed-loop system is asymptotically stable in absence of packet dropouts?
- Assume now that $K = -2/3$. Also, assume that the NCS is subject to constant delay $\tau = \log_e 1.1$ s and to packet dropouts. In particular, variable θ_k has the following periodic behavior:

k	1	2	3	4	...
θ_k	1	0	1	0	...

Is the corresponding NCS asymptotically stable?

2) Define in details the **delays in the network channel**, commenting the equations that allow to define the **propagation and frame time**. What are the typical bit rates/message sizes/frame times of the most **widespread control networks** (e.g., Ethernet, Devicenet, etc)?

3) Given a linear time invariant system

$$\dot{x}(t) = Ax(t) + Bu(t)$$

with an **admissible discrete control set** $\mathcal{U} = \{\bar{u}_1, \dots, \bar{u}_N\}$. Answer the following questions.

- Define the **boundability** of this system.
- Comment on the relationship between the concepts of **boundability** and **stabilizability**.
- Consider the system

$$\begin{cases} \dot{x}_1(t) = -3x_1(t) + u(t) \\ \dot{x}_2(t) = x_1(t) + x_2(t) + 2u(t) \end{cases}$$

Is it stabilizable? In case a digital quantized controller is used to control it with sampling time $h = 0.2$ s, what is the minimum number of bits N_B encoding the control input required to make the system boundable?